

PATIENT MISSED APPOINTMENTS DASHBOARD

PROJECT OBJECTIVE

This project aims to analyse patient appointment data to identify key factors contributing to missed appointments (no shows). The insights generated will help healthcare providers improve attendance rates, reduce resource wastage and enhance overall performance efficiency.

KEY FINDINGS

WAITING TIME HAS THE STRONGEST IMPACT ON MISSED APPOINTMENTS

Patients scheduled within 7 days have a missed rate of 14%, while for those waiting more than a week it rises to over 30%.

YOUNGER PATIENTS ARE MORE LIKELY TO MISS APPOINTMENTS

Patients in the 0-17 and 18-34 age group have the highest missed rate (22% and 24%, respectively), while senior patients (65+) have the lowest rate (16%).

MISSED APPOINTMENTS RATES REMAIN FAIRLY CONSISTENT THROUGHOUT THE WEEK

Rates range from 19% to 23%, with the lowest on Thursday and the highest on Saturday.

SMS REMINDERS APPEAR TO BE TARGETED TOWARDS HIGH-RISK PATIENTS

Patients who receive SMS reminders have an overall higher rate of missed appointments, suggesting reminders may be used for patients who are already more likely to miss appointments.

SMS REMINDERS APPEAR TO BE MORE EFFECTIVELY FOR PATIENTS WITH LONGER WAITING TIME

Patients with a short waiting time who also received SMS reminders had double the rate of missed appointments compared to those who didn't receive an SMS. However, for longer waiting times, SMS reminders were associated with a significant decrease in missed appointments. This suggests that SMS reminders may be more effective for patients with longer waiting time.

TOTAL APPOINTMENTS

110,496

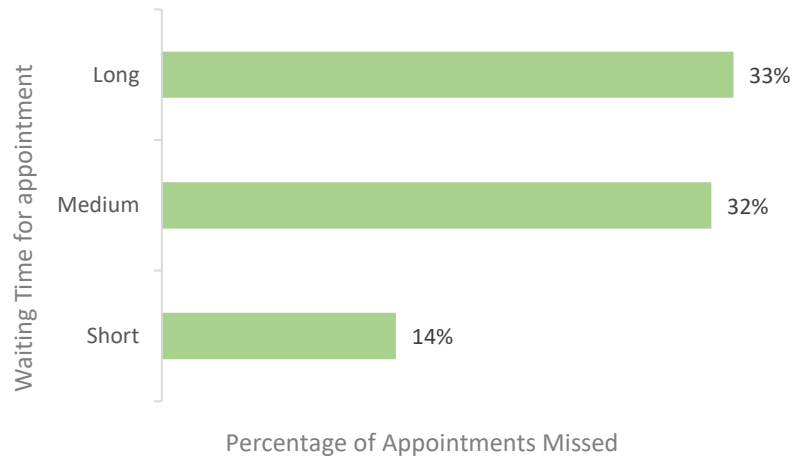
MISSED APPOINTMENTS

22,311

MISSED APPOINTMENT RATE

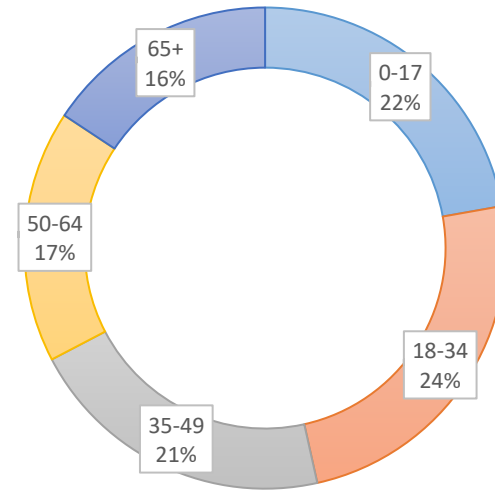
20%

Impact of Waiting Time on Missed Appointments



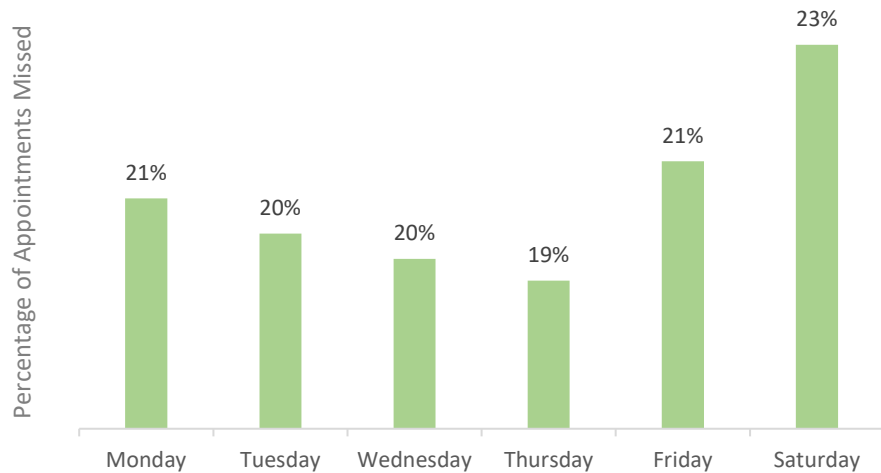
Appointments scheduled within 0–7 days have a 14% missed rate. With longer waiting times, the missed appointment rate roughly doubles.

Missed Appointments by Age Group



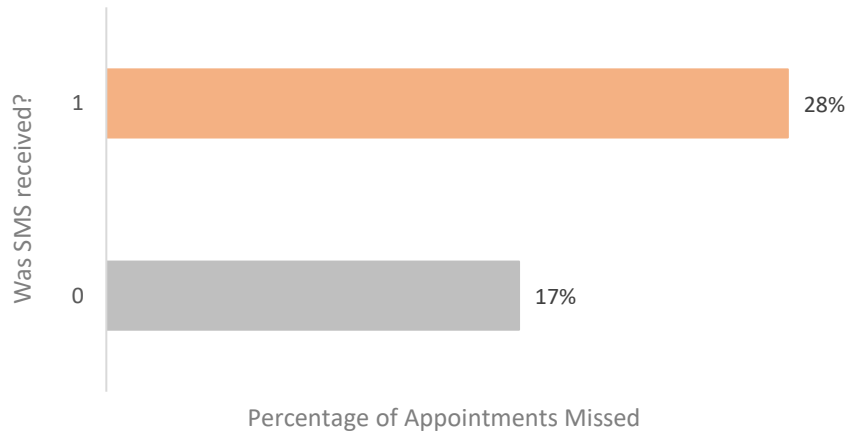
Patients aged 18–34 have the highest missed rate at 24%, while patients aged 50–64 and 65+ have the lowest at 17% and 16%, respectively.

Missed Appointments by Days of Week



The percentage of missed appointments is fairly consistent across the week, ranging from 19% to 23%. The lowest rate occurs on Thursday (19%), while the highest is observed on Saturday (23%). No appointments are recorded for Sunday in the dataset, suggesting that hospitals and clinics may not operate on that day.

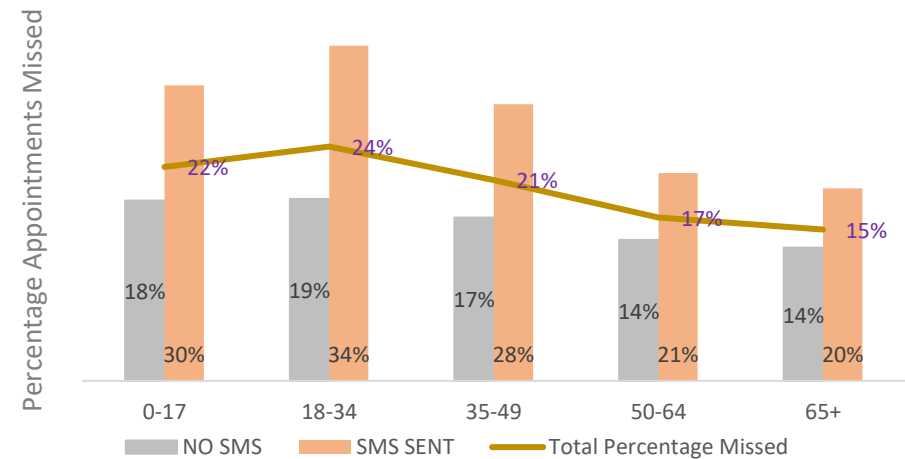
Missed Appointment Rates By SMS Received



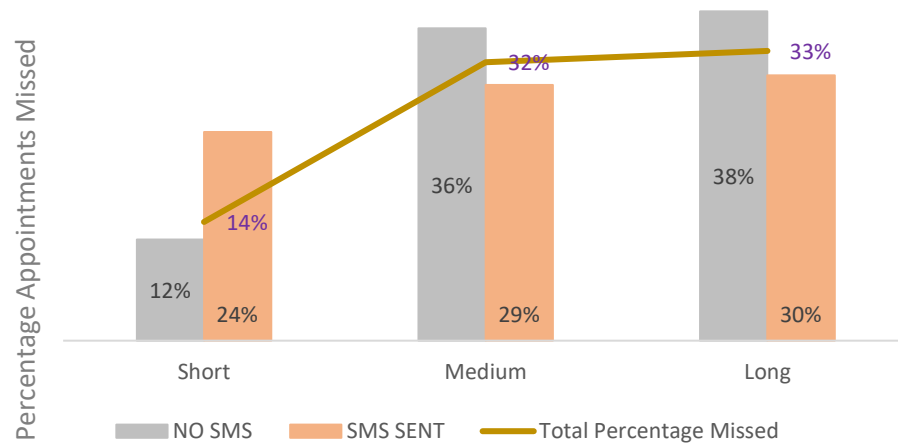
Suprisingly, patients who received an SMS reminder have a higher missed appointment rate (28%) compared to those who didn't receive a reminder (17%). This may suggest that sms reminders are targeted towards patients at higher risk of missing appointments.

Across all age groups, patients who received an SMS reminder have a higher missed appointment rate. The highest rates are observed among the 18-34 age group (34%) and the 0-17 age group (30%), while and lowest rates occur among patients aged 50-64 (21%) and 65+ (20%). This may suggest that SMS reminders are more beneficial for older patients.

Appointments Missed by Age Group and SMS Received



Appointments Missed by Waiting Time and SMS Received



Missed appointments increase as waiting time increases. However, for short waiting times, patients who received an SMS reminder have a higher missed appointment rate (24%). For longer waiting times (Medium and Long), patients who did not receive an SMS reminder have higher missed rates (36% and 38%, respectively). This suggests that reminders may be more beneficial for appointments scheduled further in advance.

RECOMMENDATIONS

- **REDUCE WAITING TIME WHERE POSSIBLE**

Since missed appointments increase significantly after 7 days, scheduling appointments within one week may help improve attendance rates.

- **PROVIDE ADDITIONAL REMINDERS FOR LONGER WAITING TIMES**

Patients with longer waiting times have higher missed appointment rates. Sending additional reminders closer to the appointment date may help reduce missed appointments.

- **TARGET REMINDER STRATEGIES TOWARD YOUNGER PATIENTS**

Patients aged 18–34 have the highest missed appointment rates. Targeted reminder strategies or follow-ups may help improve attendance among this group.

- **REVIEW WEEKEND SCHEDULING PATTERNS**

Missed appointment rates are slightly higher on Saturdays. Clinics may consider additional reminder strategies or minor schedule adjustments for weekend appointments.